Algebra Monsoon 2026 Lecture 12 Normal subgroups, quotients

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Normal subgroups, quotients

Let G be a group and $H \subset G$ a subgroup.

We have seen that H defines two partitions $H\backslash G$ and G/H of G.

Q When is $H\backslash G = G/H$?

Q When $Hg = gH$ for every $g \in G$?

Q When is $H = \ker(\rho)$ for some morphism of groups $\rho : G \longrightarrow G'$?

Q When is the equivalence relation on G corresponding to the partition $H\backslash G$ (resp. G/H) a congruence?

It turns out there is a single condition on H making all these things happen.

Def We say that H is normal (or distinguished) in G if

$$gHg^{-1} \subset H \text{ for every } g \in G.$$

Exercise: If $f: G \to G'$ is a morphism of groups and $H' \subset G'$ is a normal subgroup, then $f^*(H')$ is normal in G.

- Show that the subgroup $Int_*(G) \subset Aut_g(G)$ is normal.

Proposition The following are equivalent:

1) H is normal in G,
2) $gHg^{-1} = H$ for every $g \in G$.

Proof 1) $\Rightarrow$ 2) If H is normal in G and $g \in G$, then $gHg^{-1} \subset H$ as well as $g^{-1}Hg \subset H$. Conjugating the second inclusion by g gives

$$\underbrace{g(g^{-1}Hg)g^{-1}}_{=H} \subset gHg^{-1},$$

hence $H = gHg^{-1}$.

2) $\Rightarrow$ 1) There is nothing to prove.

Proposition The following are equivalent:

1) $gHg^{-1} = H$ for every $g \in G$
2) $gH = Hg$
3) $G/H = G\backslash H$.

Proof 1) $\Leftrightarrow$ 2) : clear.

2) $\Rightarrow$ 3) : clear.

3) $\Rightarrow$ 2) : If $G/H = H\backslash G$, then for every $g \in G$ there is a $g' \in G$ such that $gH = Hg'$. In particular, $g \in Hg'$. It follows that $Hg' = Hg$ and finally $gH = Hg$. □

Exercise: The intersection of any family of normal subgroups is a normal subgroup.

Proposition The following are equivalent:

1) H is normal in G

2) $H = \ker(\rho)$ for some morphism of groups $\rho : G \longrightarrow G'$.

Proof 2) $\Longrightarrow$ 1). Suppose $H = \ker(\rho)$ and let $g \in G$, $h \in H$. Then

$$\begin{aligned}\rho(ghg^{-1}) &= \rho(g)\rho(h)\rho(g)^{-1} \\ &= \rho(g)\, e\, \rho(g)^{-1} \\ &= e,\end{aligned}$$

so $ghg^{-1} \in H$, showing that H is normal in G.

1) $\Longrightarrow$ 2) : Suppose H is normal in G, and let $\rho : G \longrightarrow G/H$ be the canonical surjection $g \longmapsto gH$.

We claim that the equivalence relation corresponding to ρ is a congruence on the magma G — it is compatible with the law of composition. Indeed, if $g_1H = g_2H$ and $h_1H = h_2H$ for some $g_1, g_2, h_1, h_2 \in G$, then

$$\begin{aligned} g_1h_1H &= g_1h_2H \\ &= g_1Hh_2 \\ &= g_2Hh_2 \\ &= g_2h_2H. \end{aligned}$$

Therefore there is a unique law of composition on G/H making ρ a morphism of magmas. But ρ is surjective, so the magma G/H is a group. As $H = \ker(\rho)$, we are done. □

Example $Z(G)$ is a normal subgroup of G.

Indeed, for $g \in G$ and $z \in Z(G)$, we have $gzg^{-1} = gg^{-1}z = z$,

$\Rightarrow g Z(G) g^{-1} \subset Z(G)$. ∎

Example Let $i_1 : G_1 \to G_1 \times G_2$ be the morphism $i_1(g) = (g, e)$. Then $i_{1*}(G_1)$ is normal in $G_1 \times G_2$. Ditto $i_{2*}(G_2)$.

Exercise Show that the subgroup $D(G)$ generated by all commutators

$$xyx^{-1}y^{-1} \quad (x, y \in G)$$

is normal in G

Write $[x, y] = xyx^{-1}y^{-1}$.

Show that $g[x, y]g^{-1} = [gxg^{-1}, gyg^{-1}]$.

Indeed, $[gxg^{-1}, gyg^{-1}]$

$= (gxg^{-1})(gyg^{-1})(gx^{-1}g^{-1})(gy^{-1}g^{-1})$

$= gxyx^{-1}y^{-1}g^{-1}$

$= g[x, y]g^{-1}$. □

Universal property of the quotient

Let G be a group, $N \subset G$ a normal

subgroup, and $\pi : G \longrightarrow G/N$ the canonical surjection.

Proposition For every morphism of groups $f : G \longrightarrow H$ such that $N \subset \ker(f)$, there exists a unique morphism of groups $\overline{f} : G/N \longrightarrow H$ such that $f = \overline{f} \circ \pi$.

Proof Let us first show that if $\pi(x) = \pi(y)$ for some $x, y \in G$, then $f(x) = f(y)$.

Indeed, $\pi(x) = \pi(y)$

$\Rightarrow \pi(xy^{-1}) = e$

$\Rightarrow xy^{-1} \in N$

$\Rightarrow xy^{-1} \in \ker(f)$

$\Rightarrow f(xy^{-1}) = e$

$\Rightarrow f(x) = f(y)$.

Therefore there is a unique mapping $\overline{f} : G/N \longrightarrow H$ such that

It remains to show

$f = \bar{f} \circ \pi$,

that $\bar{f}$ is a morphism of groups, for which it is enough to show that

$$\bar{f}(ab) = \bar{f}(a)\bar{f}(b)$$

for all $a, b \in G/H$. Indeed, we have $a = \pi(g)$ and $b = \pi(h)$ for some $g, h \in G$, and

$$\begin{aligned} \bar{f}(ab) &= \bar{f}(\pi(g)\pi(h)) \\ &= \bar{f}(\pi(gh)) \\ &= f(gh) \\ &= f(g)f(h) \\ &= \bar{f}(a)\bar{f}(b). \end{aligned}$$

□

Proposition A morphism of groups $f : G \longrightarrow H$ is injective if and only if $\ker(f) = \{e\}$.

Proof: Clearly, if f is injective, then $\ker(f) = \{e\}$. Conversely,

Suppose that $\ker(f) = \{e\}$ and $f(x) = f(y)$ for some $x, y \in G$.
Then $f(xy^{-1}) = e$, so $xy^{-1} \in \ker(f)$ and therefore $xy^{-1} = e$. Hence $x = y$. □

Let G be a group, $N \subset G$ a normal subgroup, $\pi : G \to G/N$ the quotient of G by N, $f : G \to H$ a morphism of groups such that $N \subset f^*(\{e\})$, and $\bar{f} : G/N \to H$ the unique morphism of groups such that $f = \bar{f} \circ \pi$.

Prop. • $\bar{f}$ is injective $\Leftrightarrow N = f^*(\{e\})$
• $\bar{f}$ is surjective $\Leftrightarrow f_*(G) = H$.

Proof Suppose that $N = f^*(\{e\})$ and let $x \in G/N$ be such that $\bar{f}(x) = e$. Write $x = \pi(a)$ for some $a \in G$. As $f(a) = \bar{f}(\pi(a)) = e$, we have $a \in f^*(\{e\})$, hence $a \in N$, hence $x = e$.

Conversely, suppose that f is injective. Then $\overline{f}^*(\{e\}) = \{[e]\}$, so $f^*(\{e\}) = \pi^*(\overline{f}^*(\{[e]\})) = N$.

Next, π is surjective, so $\overline{f}$ is surjective if and only if $f = \overline{f} \circ \pi$ is surjective. $\square$